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On Multi-Agent Cognitive Cooperation: Can virtual agents behave like humans?

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ABSTRACT

Individuals tend to cooperate or collaborate to reach a common goal when the going gets tough creating a common frame of reference that is a common mental representation of the situation. Information exchange among people is fundamental for building a shared strategy through the grounding process that exploits different communication channels like vision, haptic, or voice. Indeed, human perception is typically multi-modal.

This work proposes a two-fold study investigating the cognitive collaboration process both among humans and virtual agents of a multi-agent reinforcement learning (MARL) system. The experiment with humans consists of an interactive virtual shared environment that uses multi-modal channels (visual and haptics) as interaction cues. Haptic feedback is fundamental for a good sense of presence and for improving the performance in completing a task. In this manuscript, an experiment, consisting of escaping a virtual maze trying to get the best score possible, is introduced. The experiment is meant to be performed in pairs, and the perceptual information is split among the participants. A custom haptic interface has been used for the interaction with the virtual environment. The machine learning case, instead, proposes two virtual agents implemented using a tabular Q-learning paradigm to control a single avatar in a 2D labyrinth, introducing a new form of MARL setting.

As it is known, it is not easy to get familiar with haptics for people that have never used it, and that if not properly transmitted, the cognitive workflow does not produce any improvements. However, the main findings of the proposed work are that haptic-driven multi-modal feedback information is a valuable means of collaboration since it allows to establish a common frame of reference between the two participants. The machine learning experiments show that even independent agents, implemented with properly designed rewards, can learn the intentions of the other participant in the same environment and collaborate to accomplish a common task.

1. Introduction

When a problem becomes too difficult for a single individual, human beings tend to collaborate or cooperate to achieve a common goal. In collaboration, people work together synchronously in coordination on a shared goal upon which they have already agreed and try to construct and maintain a shared conception of a problem. In cooperation, people work together while adjusting each one's selfish goal to achieve a common purpose by splitting labor and duties among participants. When individuals work together (collaborating or cooperating), they build a common frame of reference (CFR) that is a common mental representation of the situation [1]. With a CFR, partners can organize their work and perform complementary actions to solve that problem. The grounding process [2] allows partners to exchange information for constructing and updating the CFR. Different communication channels (vision, haptic, or voice) are used to update the CFR depending on the specific task. The grounding process is fundamental in scenarios where only some participants have access to particular information necessary for task completion.

This work proposes a two-fold collaborative cognitive experimental study. The first investigation takes place in an on-purpose virtual shared environment for a purely dynamic and cognitive human task using haptic-based cross-modality feedback. The other experiment tries to mimic human behavior in a simpler synthetic scenario where participants are replaced by virtual agents controlled with a reinforcement learning algorithm.

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Human collaboration has not been studied as well as cooperation in animal kingdoms like ants or bees. Cognitive collaboration operates unconsciously without the need for learning, and it is considered useful for tasks that exceed the ability of a single individual [3]. In the same way, multi-agent reinforcement learning has not been studied as deep as single agent-reinforcement learning [4]. The experiments in this work are modeled as a game that necessarily requires two users to collaborate to complete a common task. The peculiarity is that each of the two participants shares the same environment but has access to complementary details necessary to finish the task and is responsible for solving a portion of the problem. However, to efficiently solve the designed task, a collaboration should emerge among the agents not to compete but to focus on the final goal. Furthermore, the actions of the two users are combined to control an avatar to achieve the highest possible performance. The experiments conducted by human users aim to study how the learning process can behave in a collaborative scenario in which each user is necessary for the other to reach the same goal under different conditions. The objective of the machine learning experiment, instead, is to investigate a novel kind of multi-agent system in which each agent, performing individually, is not fully autonomous since it has not a complete vision of the scene and needs the other one to properly command the avatar.

Summarizing the presented work proposes the following contribution:

- a new human case study concerning the interaction among two players in a virtual environment that, by means of haptic devices and cognitive collaboration processes, have to escape a maze with complementary information at their disposal to achieve the task;
- a new Multi-Agent Reinforcement Learning (MARL) model inspired by the human experiment, in which two virtual agents that control the same avatar try to escape a given maze. The novelty, rather than the split of data that is common in many MARL systems like the partially observable environments [5], is in the fact that the avatar cannot maximize the score function without the joint decision of both virtual agents having different but complementary perceptions of the same environment.

Section 2 presents the works related to multi-modal feedback interaction and discusses multi-agent reinforcement learning. Section 3 starts describing the system and then is split into two parts: one concerning the human experiment and the other the machine learning one. Section 4 discusses the obtained results in both experiments. The conclusions, in Section 5, summarize the proposed work.

2. Related Works and Background

Theory of Mind [6] is the term used in psychology to establish that a living being has the capability of understanding that others have intentions, beliefs, desires, and perspectives that differ from their own. Machine Theory of Mind [7], instead, is used to define the type of models designed to model behaviors of other virtual agents. The presented work helps to explore both the theories. A human experiment allows investigating the capability of cognitive collaboration: a human is able to understand the intentions of another human by perceiving the behavior of a shared avatar. In the same manner, a Multi-Agent Reinforcement Learning (MARL) experiment allowed exploring the possibility of a virtual agent to learn the intention of another independent agent.

Virtual environments (VEs) may give people the impression of being in a real situation. Thanks to the improvement of technology in the areas of computer graphics and audiovisual, VEs can produce a great illusion of a real environment for the visual and auditory senses. Unfortunately, this illusion breaks when users do not experience the physical sensation of their actions. Consequently, the sense of presence in VEs disappears with the absence of haptic feedback. Haptic feedback is fundamental both for a good sense of presence and immersion because in the physical world reaction forces are available [8].

In human life, perception is multi-modal: several senses at the same time enable us to perceive sensory information from the outside [9]. In such a way, different inputs that are not available from a single modality can be exploited. In particular, vision and haptic channels can collect complimentary information like color for vision or hardness for haptics or redundant data like shape, size, or texture of an object [10]. However, it is worth noticing that such channels process information in different ways. The haptics system typically gets information in sequence, for example, after hand movements, while vision scans the scene globally [11]. It happens that inputs from one of the two channels can compensate from sensing deprecation in the other [12], but when the two modalities conflict, vision generally dominates [13]. Furthermore, unlike the visual feedback that is unidirectional, haptic interfaces establish bidirectional communication between the user and the computer. The user changes the position of the device or applies forces on it, and the controller sends information to the computing unit. The computer updates the scene accordingly and gives back the forces and the positions.

In most cases, vision and audio channels are sufficient for performing collaboration among multiple users, but there are situations in which other channels, like haptic feedback, need to be involved. In these years, several experiments have been conducted on the use of multi-modal feedback, especially vision and haptic, in different tasks [14,15]. These experiments have shown to obtain higher performance using both feedback information. However, haptic feedback has usually been used just for augmentation or redundancy rather than as a substitute for visual information or the primary sense.

The research in the fields of rendering techniques and haptic interfaces has progressed a lot in the last few years. Results have attested the important role that haptics has in the sensorial experience of the users. There is a strong belief that haptic feedback in VR could provide sufficient realism for replicating real word interactions. Haptic feedback has been widely adopted in [16,17] where it increases the salience of information providing new cues. [18] shows how haptic feedback can even improve social presence in shared virtual environments (SVEs). Concerning teleoperation, the performance increases significantly when haptic feedback is provided during the telemanipulation of remote objects [19]. Moreover, expert surgeons state that haptic feedback is necessary for surgery and that training using VR simulations is inefficient due to the lack of realism given by the absence of haptic feedback [20]. Haptic cooperative environments have gained great attention in recent years, especially for mini-invasive surgery (MIS) [21].

All the cited systems use force feedback as means of realism or redundancy and not for guidance in collaboration. The proposed work, instead, uses haptics as the main tool to drive the movement of a virtual agent. The haptic feedback is the principal means of interaction between the two participants for cognitive collaboration in accomplishing a given task. The result of the interaction of the partners affects only the visual channel by design choice. Further details on the experiments are presented in Section 3.1.

All agents in a collaborative multi-agent system can influence each other. This is also true in MARL systems in which it is relevant to ensure that the actions selected by each agent result in the optimal decision for the whole system. Most of the successful reinforcement learning (RL) applications like chess[22], Go [23], robotics [24], and autonomous driving [25], involve more than one single agent. Therefore, they are categorized as multi-agent RL, which has a quite long history and has recently been back in vogue thanks to the development of single-agent RL techniques. MARL can be described as a sequential decision-making problem of multiple agents that work in a shared environment. Each agent optimizes its long-term return while interacting with the environment and the other agents [26]. MARL strategies have been applied in different applications filed in engineering systems like robotics [27], intelligent transportation systems [28], smart grid [29], sensor network [30]. Many techniques have been applied depending on the nature of the problem like minimax deep deterministic policy gradient [31], mean-field theory [32], win or learn fast proximal policy [33], deep recurrent q-learning [5], and others. The following aims to summarize a necessary background on single-agent RL and MARL.

An RL agent is designed to perform sequential decision-making interacting with an environment, which is formulated as a Markov decision process (MDP).

Definition 1 (***Markov Decision Process***). A Markov decision process is defined by a tuple (S, A, P, R), where S is the state space; A is the action space; $P_a(s, s\prime) : S \times A \times S \rightarrow \mathbb{R}$ represents the transition probability from any state $s \in S$ to a new state $s\prime \in S$ given any action $a \in A; R_a(s, s\prime) : S \times A \times S \rightarrow \mathbb{R}$ denotes the reward function that gives the immediate reward to an agent after a transition from the state s into state $s\prime$ performing the action a.

The main property of such an MDP is that the individual reward and the current state depend only on the previous state and the taken action, and satisfies the stationarity assumption.

At each time t, being the agent in state s_t, it chooses an action a_t to execute and move to state s_{t+1} with a transition probability $P_{a_t}(s_t, s_{t+1}) = Pr(s_{t+1}|s_t, a_t)$ and receives an instantaneous reward $R_{a_t}(s_t, s_{t+1})$. The objective of an MDP is to find a policy $\pi : S \rightarrow A$, which is a rule to decide which action to take in each of the possible states with the aim of maximizing the discount accumulated reward

$$\sum_{t \geqslant 0} \gamma^t R_{a_t}(s_t, s_{t+1}), \text{ where } a_t = \pi(s_t), \tag{1}$$

where $\gamma \in [0, 1]$ is a discount factor acting as a tradeoff between the instantaneous and the future rewards.

The state-action function $Q_\pi(s, a)$, also known as , and the value function $V_\pi(s)$ are defined. The Q-function contains for each state $s \in S$ the chosen action $a \in A$ under the policy π, and the value function maintains a real number for each state $s \in S$ under the policy π. Reinforcement learning aims to discover the optimal policy π^* without knowing the model. The agent learns the policy by trial and error in interacting with the environment.

Model-free RL algorithms can be subdivided into two main categories:

- value-based methods that aim at identifying a good estimate of the state-action value function Q_{π^*}. The Q-learning [34] is one of the most popular algorithms. It will be presented in Section 3;
- policy-based methods that search over the policy space. The most used approach is the policy gradient method [35] that updates the parameter along the gradient direction.

MARL systems also cope with sequentialproblems but with more than one agent. Therefore, the evolution of the system and the reward of each agent are influenced by the joint actions of all of them, thus invalidating the stationarity assumption [36]. As consequence, the long-term reward of each agent that should be optimized is a function of the policy of all the other participants. MARL systems can be subdivided into two categories: Markov game [37], which is explained in the following, and extensive-form game [38], which is out of the scope of this work.

Definition 2 (***Markov game***). A Markov game is defined by a tuple $(N, S, \{A^i\}_{i \in N}, P, \{R^i\}_{i \in N})$, where $N = 1, \ldots, n$ is the set of agents, S is the state space observed by all agents, A^i is the action space of agent i. If A is the cartesian product of the action of each agent $(A := A^1 \times A^2 \times \ldots \times A^n)$, then $P_a(s, s\prime) : S \times A \times S \rightarrow \mathbb{R}$ represents the transition probability of going to any state $s\prime \in S$, being in any state $s \in S$ performing any joint action $a \in A; R^i_a(s, s\prime) : S \times A \times S \rightarrow \mathbb{R}$ denotes the reward function that gives the immediate reward to each agent i, given a transition from s to $s\prime$ performing action a.

At each time t, each agent $i \in N$ performs an action $a^i_t \in A^i$ depending on the state of the system s_t. The system goes into states s_{t+1} and rewards each agent according to $R^i_{a_t}(s_t, s_{t+1})$, where $a_t \in A$. Each agent aims at optimizing its own long-term reward with the policy function $\pi^i : S \rightarrow A^i$. The value function $V^i : S \rightarrow \mathbb{R}$ of each agent i depends on the joint policy $\pi : S \rightarrow A$, where $\pi := \prod_{i \in N} \pi^i(s)$. Therefore, the solution of a Markov Game deviates from the one of a Markov Decision Process since the optimal performance of each agent depends also by all the other members.

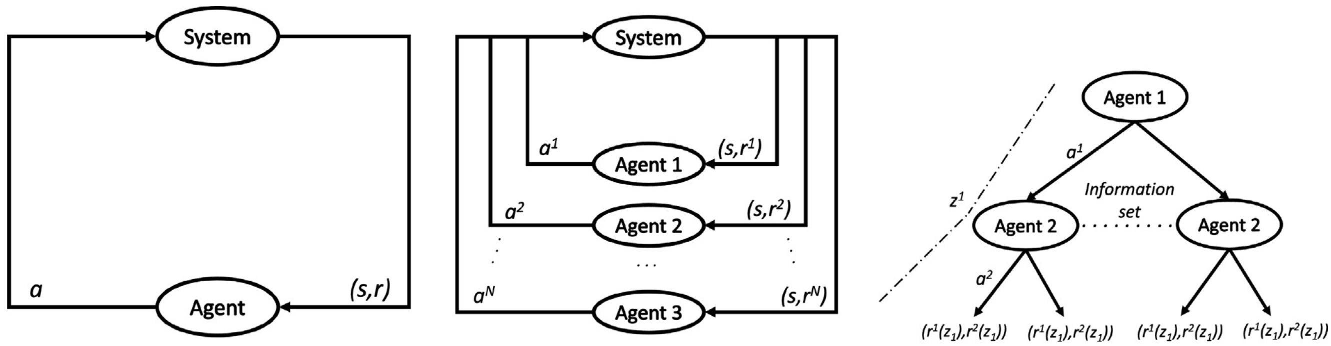
Fig. 1 illustrates schematically an MDP under different settings.

Since the objectives of all agents are not , the learning goals in MARL are multidimensional. This led to the challenge of coping with equilibrium points and, since all agents update their policies according to their interests concurrently, the environment becomes no more stationary, thus, invalidating the most theoretical analyses in the single-agent settings. Moreover, the joint action space increases exponentially with the number of agents causing scalability issues [39]. Finally, each agent can have limited access to the observations of the others, leading to possible suboptimal decision rules locally. Fig. 2 shows the three information structures used in MARL Literature. In the first case [40], it exists a central controller that acts as a supervisor aggregating information from the agents like joint rewards, joint actions, joint observations, and even design policies for all agents. The other two are decentralized. In particular, in the second scenario [41], the agents can communicate, for example, using the network, with each other to exchange information. The last setting is fully decentralized [30], and there is no explicit communication among the agents. However, the agents make decisions based on their local observations that could contain some global information, without any communication or coordination of a central unit.

In Section 3.2, this work proposes a decentralized MARL system of two not homogeneous agents that have different reward functions and have access to two complementary sets of information. The two agents form a team to accomplish a common task while maximizing their own reward, given the information at their disposal. Real-world applications in which this new paradigm is applicable are diverse. An immediate analogy is a single physical robot equipped with multiple sensors that perceive the world with different information: the sensors are the agents that control the avatar, which is the robot. The sensors need to learn how to collaborate to perform the best feasible action for the robot since the local optimum of the single unit is not always the global optimum for the robot.

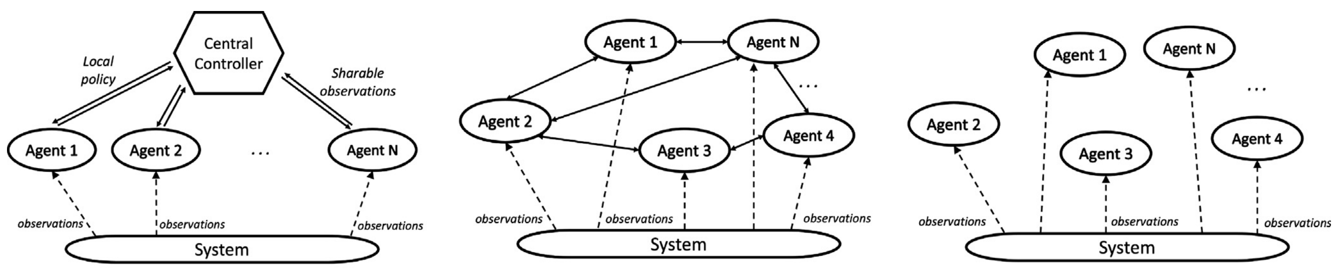
3. System Description

The following section describes first the setup of the experiment conducted by humans in a shared virtual environment using vision and haptic feedback as interaction modalities, and then



(a) Naive MDP (b) Markov game (c) Extensive-form game

Fig. 1. Markov Decision Process schematic diagrams.



(a) Centralized settings (b) Decentralized setting with networked agents (c) Fully decentralized setting

Fig. 2. Multi-Agent Reinforcement Learning Settings.

explains the new decentralized multi-agent reinforcement learning design in a simple scenario.

3.1. Human Experiment

One of the most important aspects related to virtual collaborative systems is the representation of users in the virtual scene. This work proposes a particular kind of interacting collaborative virtual environment in which the actions of two users affect the motion of a unique avatar. The two players have to cooperate to complete a 3D maze escape task. The visual information of the virtual scenario (A) is distributed among them in two subsets (A_1 and A_2). In particular, one of the players can see the wall of the maze and the final destination (target to reach), while the other can locate the bonuses (stars) and the maluses (bombs), as depicted in Fig. 3.

The visual channel is beneficial for the users to understand and predict the intended direction of movements of the other participant, and to guess possible environment constraints and the final destination location. The users interact with the virtual scenario by means of a haptic interface that determines the desired velocity vector applied to the moving agent.

Fig. 4 shows the experimental setup. The users can not see each other, but can perceive each other through the visual scene update due to haptic interaction.

The goal of the collaboration is to achieve the best score in reaching the final destination. The score is computed as:

$$score = 10 * n_{stars} - 15 * n_{bombs} + 20 * (max_{time} - t). \qquad (2)$$

The score depends on the difference between a chosen maximum time limit (max_{time}) and the overall task execution time, on the number of collected stars (n_{stars}) and on the number of collided bombs (n_{bombs}). In the experiments, max_{time} was selected as 120 seconds.

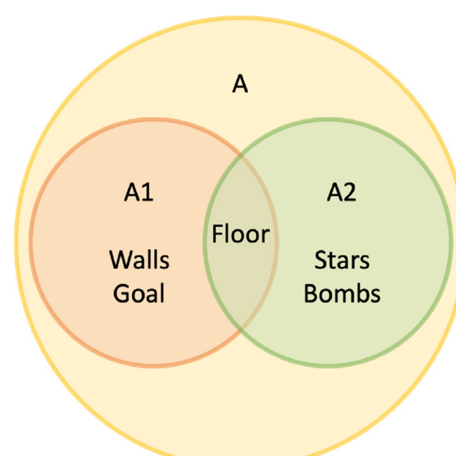


Fig. 3. Virtual Environment information sets.

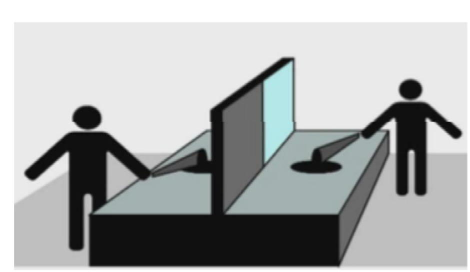


Fig. 4. Experimental setup.

Fig. 5 shows how the maze visual information is split among the users on a frame of the experiment. The virtual maze is composed

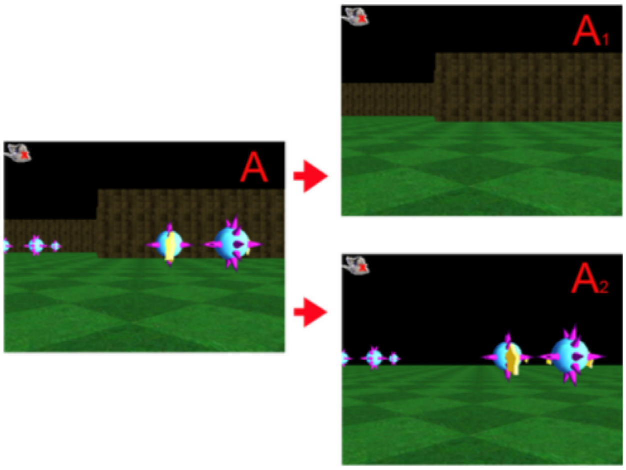


Fig. 5. Virtual maze seen by the two participants.

of 300 x 700 tiles, and it contains 15 stars and 20 bombs dislocated all around the scenario. The starting point and the goal are not far from each other. Thus, if the two users manage to control the haptic device and the cognitive cooperation process, they would not take too much time to achieve the goal. On the contrary, if they miss the task, they can be lost in the maze, and it would take time to recover the path. Such information can be seen in the sketch representation of the scenario in Fig. 8. The objective of the two players is to escape the maze by creating a path from the starting position to the destination goal in the shortest time possible while trying to collect the highest number of stars and avoiding the bombs along the path.

A movement in the virtual world corresponds to the combination of the relative motions induced by the users' haptic interfaces as shown in Fig. 6. The device returns force feedbacks proportional to the generated spatial displacement with respect to its rest position (center of the haptic interface workspace). In this way it is possible to use a velocity control paradigm and explore a large virtual environment with the virtual agent. Therefore, what is interesting to investigate is the mental bridge that the designed experiment builds in the users mind. If a user wants to proceed in a specific direction, but the resultant movement of the virtual scene does not match the desired one, it means that the two users are not coordinating properly or that they are hitting a wall (information that just one of them holds).

The experiment has been designed on a distributed framework that allows developing multi-modal real-time and interactive experiment in a virtual shared environment. It can manage visual, auditory, and haptic feedback giving a strong sense of immersion and presence to the users. However, in this experiment, only visual and haptic channels are adopted. Even if many common tasks have visual sensing as primary mode, haptic is fundamental for controlling dynamic systems like the one that is proposed in the experiment. In dynamic systems, indeed, to reach the desired state, an operator should give input to the system taking into account the dynamic response but also its history. Mechanical properties of the system like damping, inertia, or stiffness, are read easily through haptic feedback rather than learned visually observing how an object responds to known forces overtime. The haptic interaction and rendering are handled using a software module that manages the interaction of the devices with the virtual scene. Two separated Virtual Machines (VMs) communicating via shared memory are implemented to solve the problem of multithreading. The main VM takes care of the graphic loop while the other is responsible for the haptic loop. A fast real-time haptic control is realized transmitting packets via UDP for guaranteeing a responsive system and a stable loop. It is worth to notice that the haptic loop is local to each of the two users. They can perceive each other just seeing how the motion of the virtual avatar (camera) is updated, but do not receive the force applied by the partner. Therefore, the haptic loop goes to 1KHz, is stable, and does not experience any latency. The only latency that could be perceived is related to the update of the graphical scene. Due to this latency, the state of the two users may be inconsistent for a fraction of second. In such a case, dead reckoning is applied. However, in the experiment, having at disposal a dedicated LAN, the latency of the UDP transmission is far lower than both the graphic update rate (30*Hz*) and the simulation rate (50*Hz*).

The haptic interface used in the experiment is the GRAB interface [42]. The device (Fig. 7) has three actuated DOF in the parallel-serial configuration 2R+1P (2 rotational and 1 Prismatic joints), plus three passive (3R) rotoidal degrees of mobility (DOM). The two orthogonal pairs work in parallel to control the orientation of a plate to which the third prismatic joint is attached. A balance weight is also placed on the rotating plate to minimize the motor efforts when the prismatic joint is fully extended. The first three joints allow moving and tracking the end-effector (EE) in a 3D space and provide continue force feedback up to 5N in the worst case condition while the peak forces are up to 20 N. The other three DOMs let the EE be oriented in any direction. The workspace shape is a square-based pyramid trunk, but only

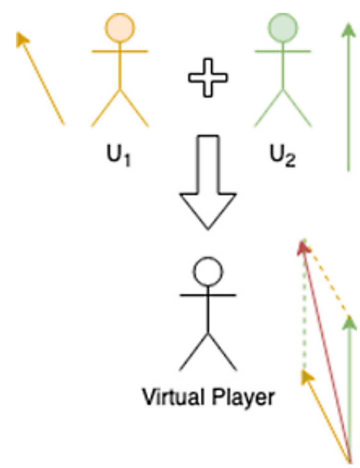


Fig. 6. Velocity vector of the virtual player.

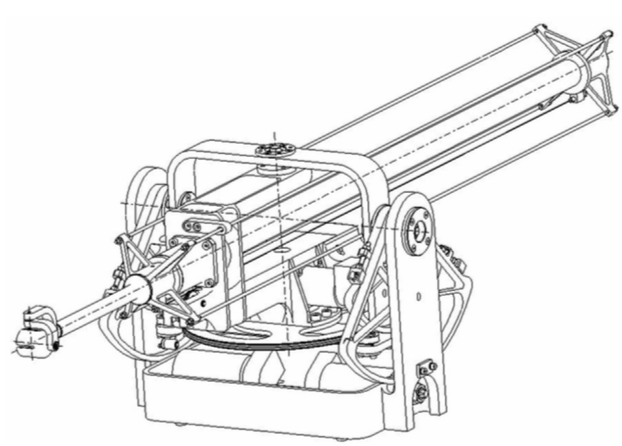


Fig. 7. Haptic device.

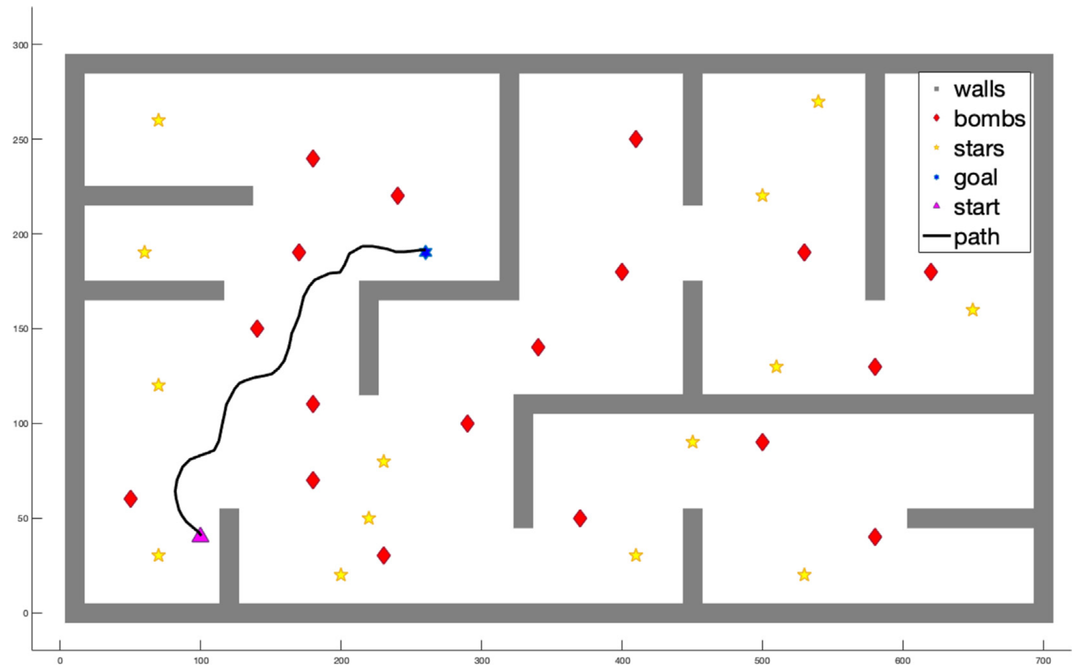


Fig. 8. Trajectory of group 4 in trial 3.

a rectangular parallelepiped of size 400x400x600 mm is used. The motor controller is equipped with encoders that allow resolving spatial position up to 100 micrometers in the worst-case condition. The device has a position accuracy at zero loads less than one percent. The controller also handles basic functionalities like gravity compensation, initial calibration, acoustic warnings, and sharp motion protection in the case of programming or communication errors. The user can experience the force feedback while orienting the hand in any direction of the space. The force feedback is calculated as a spring-dumper attached to the camera. Therefore, the user should apply a force proportional to the desired displacement to move or rotate the camera.

Ten volunteers, from 22 to 31 years, participated to the experiments for evaluating the cognitive cooperation in a shared virtual environment escaping a virtual maze using visual and haptic feedback. The participants were eight males and two females, mostly students with previous computer experience. Fig. 8 depicts the labyrinth that does not change during experimental sessions. The picture shows the configuration of the maze representing the walls in gray blocks, the bombs as red diamonds and the stars as yellow stars. The starting point is shown as a magenta triangle and the target goal as a blue filled star.

The participants were divided into five pairs, and none of them have experienced this system before. Each unit of the pair can perceive visually specific details of the scene. A collaboration between the users is needed to escape the maze achieving a high score and we define U_1 the first user and U_2 the second user. Each group, after a 15 minutes practice on a different scenario from the one of the core experiment, performed four trials in which changes have been adopted as described in the following:

1. trial T1: $(U1|A1)|(U2|A2)$. It is the reference trial where every user can see complementary details of the scene;
2. trial T2: $(U1|A)|(U2|A)$. Both of the users can see the overall environment. The objective of this trial is to let experience the shape of the virtual maze;
3. trial T3: $(U1|A1)|(U2|A2)$. It has the same condition as the first trial since it is used for assessment;
4. trial T4: $(U1|A2)|(U2|A1)$. The role of the two users has been swapped to investigate if visual memory influences the performance of the task or if the previously gained knowledge drives the interaction even with the visual channel only.

It has been decided to repeat the experiment four times changing users knowledge because it would be hard for participants, during the first trial, to get confident with the cognitive cooperation driven by haptic feedback and partial visual information. When the virtual scene does not update as desired, they should speculate if the companion user is applying a force with the same magnitude, but opposite direction or they are hitting an obstacle.

Furthermore, to investigate the influence of the cognitive flow in achieving better performance during the subsequent trials, groups one and three were asked to explain the task and possible strategies to the following groups (groups two and four, respectively) after the end of their trial session. Therefore, groups one, three, and five played the game without any advice, while groups two and four knew what to expect before starting.

For each group, after the experimental session, the members of the pair were asked individually to complete a questionnaire. They were invited to report:

- if the haptic feedback has been useful;
- the easiness of the task;
- the subjective estimation of the time needed to complete the task;
- the adopted strategy used for each trial;
- when applicable, if the advice of the previous group has been useful for building a strategy.

3.2. Multi-Agent Reinforcement Learning

The machine learning experiment proposes a new MARL model, depicted in Fig. 9. Two virtual agents implemented as tabular Q-learning perceive different and complementary cues of the environment necessary to an avatar to accomplish a certain task. Therefore, seeing different information could happen with a high

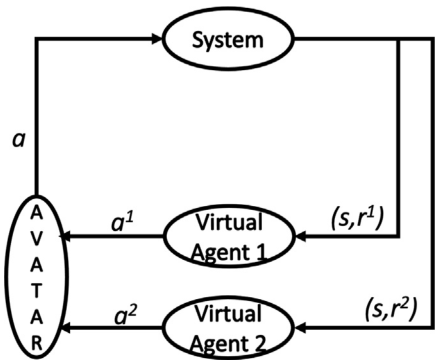


Fig. 9. Proposed MARL model.

probability that the agents perform different actions from each other. The combination of the two actions determines the behavior of the avatar in the environment. According to the movement of the avatar, the two virtual agents receive two different rewards. In particular, a simple maze scenario has been chosen to test this new MARL model, trying to resemble the same condition of the human experiment discussed in the previous section but reproduced in a discretized 2D space. In particular, the avatar moves in a confined maze with a wall, two bombs, and a goal, like in Fig. 10. Virtual agent A can see the wall and the goal, while virtual agent B can see the bombs, like the human experiment. Each virtual agent can move only up, down, left, and right of one cell in the grid maze, while the avatar moves depending on the combination of $action_A$ and $action_B$. Three different policies of action combination have been analyzed. Since one agent can only perceive walls, it is intended to find out if it can learn to avoid them and head towards the goal while the other agent learns to avoid bombs. The aim is to find out if both agents can learn the way to reach the goal unharmed.

The following definition presents in a more formal and general way the proposed setting described above.

Definition 3. The proposed model is defined by a tuple $(N, S, \Omega, O, \{A^i\}_{i\in N}, A_v, P, \{R^i\}_{i\in N})$, where $N = 1, \ldots, n$ is the set of *virtual agents*, S is the state space, Ω is the set of observations, $O^i \subset \Omega$, is the observation space of agent i, where $\bigcup_{i=1}^{N} O^i = \Omega$, and A^i is the action space of agent *i*. If A is the cartesian product of the action of each agent ($A := A^1 \times A^2 \times \ldots \times A^n$), then $A_v \subset A$ contains only the set of feasible actions the single controlled *avatar* can do. Therefore, $P_a(s, s\prime) : S \times A_v \times S \to \mathbb{R}$ represents the transition probability of going to any state $s\prime \in S$, being in any state $s \in S$ performing any joint action $a_v \in A_v$; $R^i_a(s, s\prime) : S \times A \times S \to \mathbb{R}$ denotes the reward function that gives the immediate reward to each *virtual agent i*, given a transition from s to $s\prime$ performing action a.

In the following, a brief explanation of the Q-learning algorithm that is used as the learning mechanism is given before discussing three policies.

3.2.1. Q-learning

Given an agent that interacts with a Markov Decision Process, the Q-learning is a model-free reinforcement learning algorithm that finds an optimal policy for maximizing the expected value of the total reward, starting from the current state, over any successive steps. Q-learning is guaranteed to find an optimal action-selection policy for any given finite MDP, under infinite exploration time, and a partly-random policy.

The algorithm uses a quality function $Q : S \times A \to \mathbb{R}$ for computing the quality of a state-action combination. In the beginning, Q is initialized to an arbitrary fixed value, and, in the proposed work, it assumes the form of a matrix (thus, tabular Q-learning). Then, at each time t, being the agent in the state s_t, it performs an action a_t, receives the reward r_t, goes in a new state s_{t+1}, and Q is updated. When the agent is in a state s_t, the best action a_t is the one that maximizes the total future reward R_t. If $Q(s_t, a_t)$ is the predicted value of the total future reward R_t for the pair (s_t, a_t), then, the best action to be performed in state s_t is the one that maximizes $Q(s_t, a_t) : a^* = argmax_a Q(S_t, a_t)$. The core of the algorithm is the Bellman equation, and the value function Q is updated using the temporal difference method as follows:

$$Q^{new} = Q^{curr}(s_t, a_t) + \alpha(r_t + \gamma max_a Q(s_{t+1}, a) - Q^{curr}(s_t, a_t)) \quad (3)$$

where, $Q^{curr}(s_t, a_t)$ is the current Q value, α is the learning rate and is $0 < \alpha < 1, r_t$ is the reward received moving into state s_{t+1} performing action a_t being in state s_t, γ is the discount factor, which is $0 < \gamma < 1$, and balances between immediate and future rewards, and $max_a Q(s_{t+1}, a_t)$ is the estimate of optimal future value. An episode of the algorithm ends when the new state is a final state.

It is worth noticing that there are three important parameters:

- α, which is the learning rate, and controls the difference between previous and new Q-values;
- γ, which is the discount factor, and balances between immediate and future rewards. If $\gamma \approx 0$, the agent is myopic, while, if $\gamma \approx 1$, the agent considers future rewards with greater weight;
- ϵ, which is the exploration factor, and represents the probability of selecting a random action. Initially, $\epsilon \approx 1$, and then it is gradually reduced to 0. In the beginning, the Q values are unknown. Therefore, the agent needs to estimate by exploration, i.e., trying random actions. By selecting a greedy action, the agent is exploiting his current knowledge encoded in the Q values, while, selecting a non-greedy action, the agent is exploring, searching for better actions to improve his estimates.

3.2.2. Experiment setting

The state space is a $S : 8 \times 8$ grid space maze, and each cell represents a state. The action space of each of the two virtual independent agents is $A : \{up, down, left, right\}$. An action can make the avatar move of only one cell. The action space of the avatar is the combination of action a_1 for the first agent and action a_2 for the second agent. The two virtual agents have different reward functions.

In the following the three policies for combining the actions of the two virtual agents are presented. The aim is to explore the behavior of the agents in a control-shared interaction with the environment.

- **Standard Reward**: Agent 1 receives a reward of -50 whenever the avatar hits a bomb. Agent 2 receives a reward of -30 whenever the avatar hits a wall. Both agents receive a negative reward of -10 for each cell traversed by the avatar. If the avatar reaches the goal, a reward of +100 is given.
- **Time-sharing**: In this configuration, each agent gets a time slot to control the avatar with its action. The time slot is switched on every update for each agent alternately. The reward received by the virtual agents is the same of the previous case.
- **Two-step reward**: After the normal reward received as in the first experiment, each agent gets a +5 reward when their actions match in the same direction.

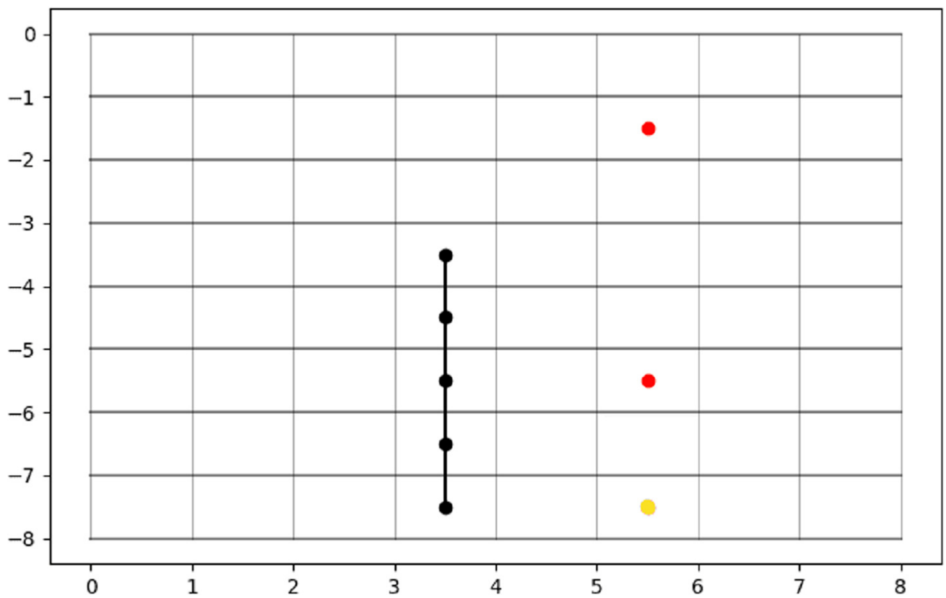


Fig. 10. 2D maze. The black spots united by a black line represents a wall that serves as an obstacle. The red spots are the bombs that constitute damaging obstacles, and the yellow spot is the Goal.

4. Discussion

4.1. Human Experiment

Analyzing the data collected for each group in the human experiment, almost all of them had some form of progress at each subsequent trial using the two feedback modalities. Only group 2 had variable results because of a participant that was very confused by the use of haptics. It should be taken into consideration that the fourth trial always had poorer performance just respect to the third one, due to the fact that the roles, and therefore the visual information, have been swapped. Indeed, trial 3 reproposes the condition of the first one but after having gained some experience in the task the participants usually achieved the best performance.

Fig. 11 depicts the trajectories followed by group 1 in the four trials . Fig. 11a shows that the first trial has a very wired path but the users should become familiar with the haptic device, the virtual environment and the the interaction among them. In the following trial (11b), in which both users can see the overall environment at the same time, the trajectory is smoother and the users explore the environment after having mastered the haptic interaction. The trial shown in (11c) reveals a nearly direct trajectory toward the goal demonstrating that the users have localized the position of the final destination and that they can handle the collaboration with the partner mediated by the haptic device. In Fig. (11d), the experimenters still reach the goal but perform a less smooth trajectory with a higher completion time due to the fact that roles have been switched. Such a result should demonstrate that the previous trials are not determined by visual memory, but instead, are due to the gained cognitive skills in the collaboration process.

Furthermore, collected data supported by the questionnaire's answers show that every group adopted a different strategy for completing the level. Some of them preferred to collect the stars avoiding the bombs, while others decided to reach the goal as soon as possible without caring about bonuses.

As mentioned in Section 3, group 2 and 4 received information about the maze from the previous groups. Only group 4 was able to exploit at best the cognitive flow of group 3. Indeed, it performs the best timing score with respect to the other groups. Fig. 8 validates this thesis showing the smoothness of the trajectory. Group 4 was also able to achieve the best coordination of the haptic interaction generating smooth force profiles.

Table 1 summarized some important information like the time spent to reach the final destination, the number of collected stars and the number of collisions with the bombs for each group in each trial.

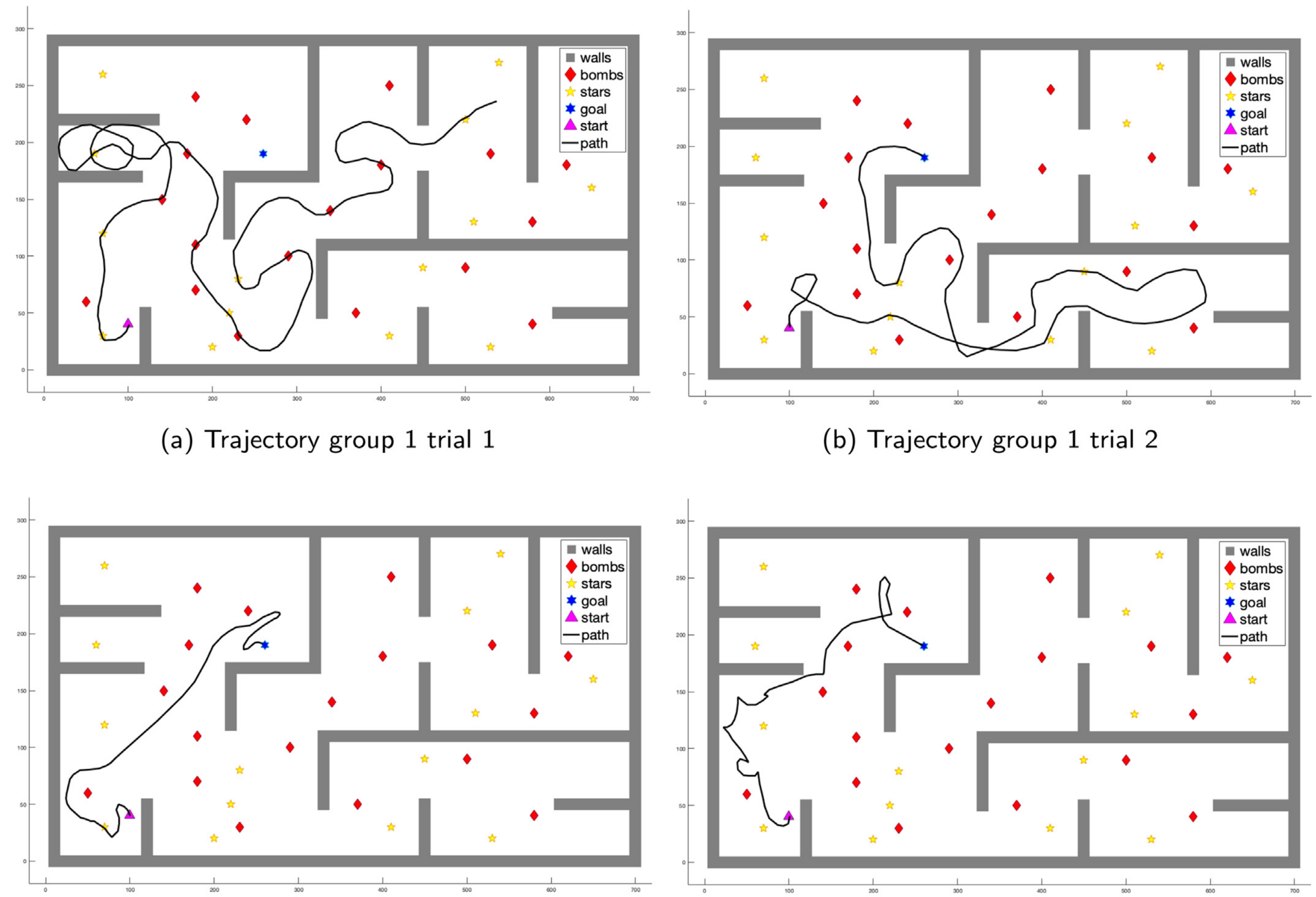
The participants reported in the questionnaires that the task was not easy to perform but that the haptic feedback was comfortable. They also stated that they changed their strategies among the trials.

4.2. Multi-Agent Reinforcement Learning

There are some considerations that are worth noticing about the three settings. Fig. 12 shows an example of the behavior of the avatar for each setting starting from the same state. In particular, in the *Standard Reward* scenario, the two agents learn the environment and are able to make the avatar reach the goal, avoiding the bombs, but the avatar performs not feasible actions in a 2D discrete space. Such a behavior emerges when the two virtual agents issue orthogonal actions (like up-right, bottom-left, etc.). Fig. 12a graphically shows the situation. Since the environment is discrete, each action a should make the avatar moves in the same unit space. Therefore, after the training, the only feasible actions for the avatar are the ones in the $A : \{up, down, left, right\}$ set, and diagonal movements are considered not feasible.

This problem disappears in the *Time Sharing* setting because there is a time slot, which passes from one agent to the other, in which each agent gets the exclusive control of the avatar for computing the next state. Fig. 12b reveals that the two agents manage to reach the goal while avoiding the bombs, but with an erratic path giving the impression that any form of collaboration is in act to optimize the path.

The last setting is the one that resembles most the human experiment thanks to the second step reward given to the agents when both of them perform the same action while perceiving different observations. This second-step reward can be seen as the mental process that a human does during the experiment of



(a) Trajectory group 1 trial 1 (b) Trajectory group 1 trial 2

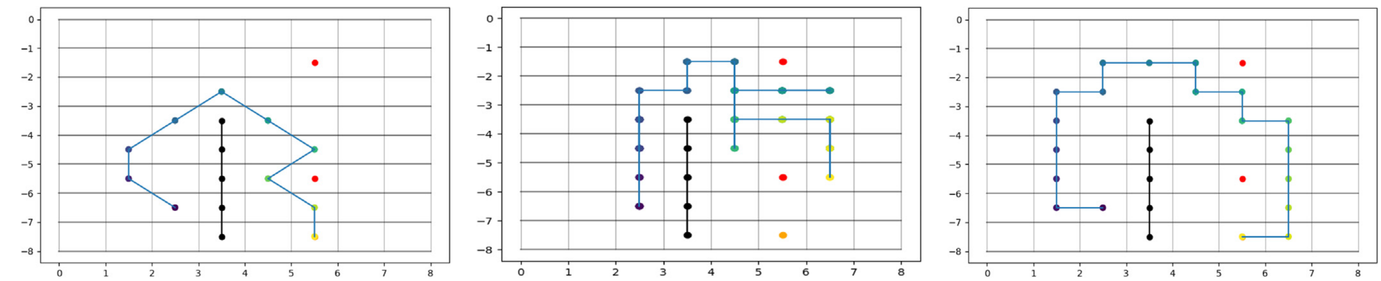
(c) Trajectory group 1 trial 3 (d) Trajectory group 1 trial 4

Fig. 11. Trajectories of group 1 in the four trials.

Table 1
Experimental sessions information.

		Time (s)	bombs	stars	score	normalized score
Group 1	T1	298.63	7	6	-3617.6	-1.42
	T2	23.42	0	4	1971.6	0.78
	T3	36.77	0	1	1674.6	0.66
	T4	51.96	0	0	1360.8	0.54
Group 2	T1	138.16	0	6	-303.2	-0.12
	T2	54.21	0	3	1345.8	0.53
	T3	287.33	17	8	-3521.6	-1.39
	T4	322.88	18	5	-4277.6	-1.68
Group 3	T1	41.42	18	2	1321.6	0.52
	T2	48.89	1	10	10507.2	0.41
	T3	25.11	2	4	1907.8	0.75
	T4	60.94	0	5	1231.2	0.48
Group 4	T1	63.59	18	10	958.2	0.38
	T2	21.07	0	0	1978.6	0.78
	T3	07.14	0	0	2257.2	0.89
	T4	36.03	18	14	1549.4	0.61
Group 5	T1	263.45	1	14	-2744	-1.08
	T2	65.22	6	12	1125.6	0.44
	T3	56.10	14	5	1118	0.44
	T4	77.63	1	4	872.4	0.34

inferring the intention of the other participant seeing how much the movement of the avatar differs from the one that he/she controlled with the haptic device. Since both the state space and the action space are discrete, and the agents perceive complementary information, the second-step reward allows the agents to build a feasible common frame of reference to collaborate properly. The two virtual agents learn a CFR to behave correctly after 50000 episodes. Fig. 13 depicts such a learning process. The first line of the figure contains the heat map matrix of the rewards of agent A1, A2, and the avatar, which is the combination of the other two,

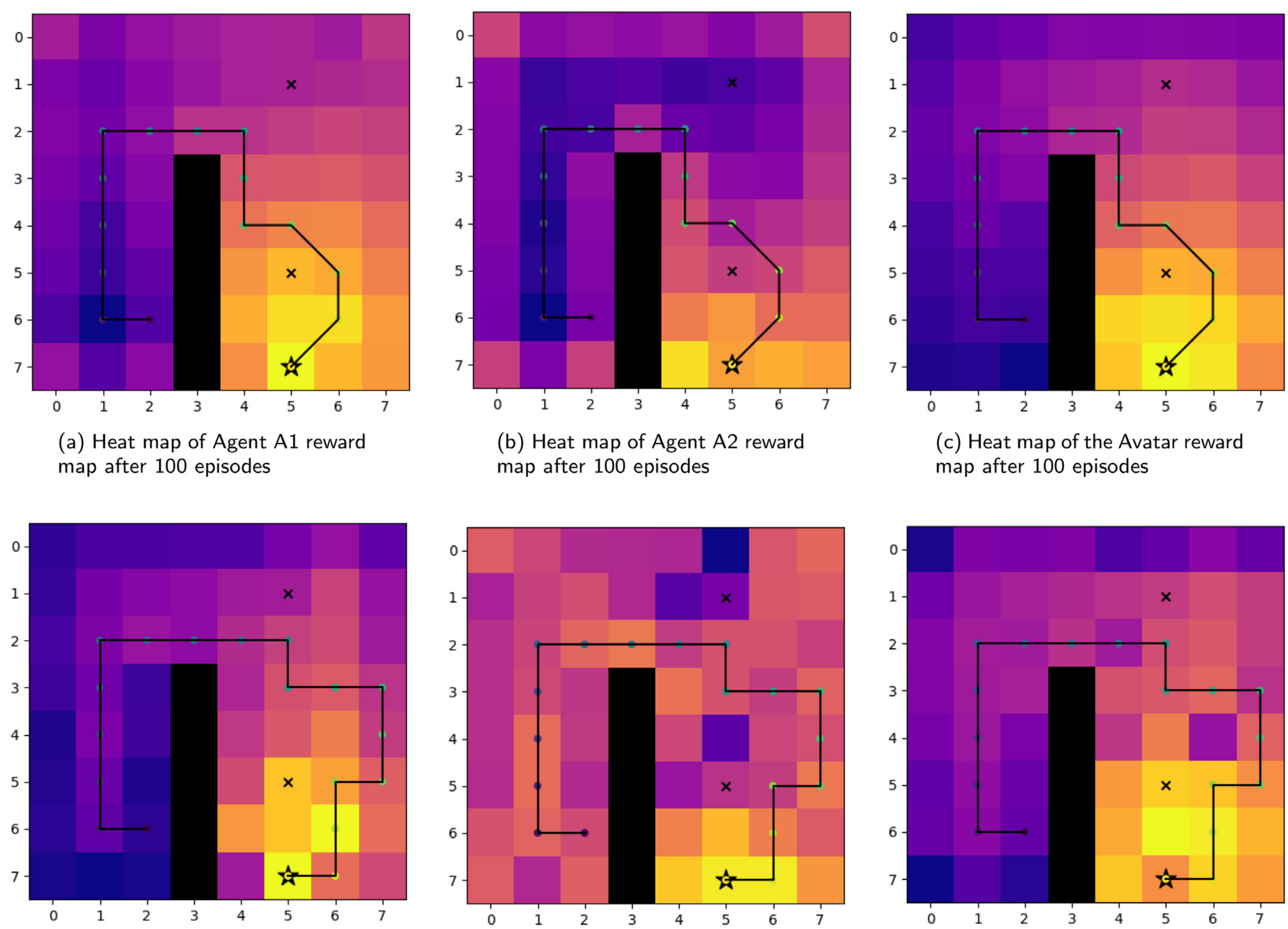


(a) Standard reward setting result

(b) Time-Sharing setting result

(c) Two Step Reward setting result

Fig. 12. Comparison of the avatar behavior in the three settings.



(a) Heat map of Agent A1 reward map after 100 episodes

(b) Heat map of Agent A2 reward map after 100 episodes

(c) Heat map of the Avatar reward map after 100 episodes

(d) Heat map of Agent A1 reward map after 50000 episodes

(e) Heat map of Agent A2 reward map after 50000 episodes

(f) Heat map of the Avatar reward map after 50000 episodes

Fig. 13. Learning process in the two-step reward setting.

respectively, after few episodes. The second line, instead, shows the same information after 50000 episodes. Darker colors in the heat map represent lower reward values, light colors, instead, up to yellow color, determine higher values. Like in the human experiment in which the players understand to cooperate throughout the haptic and visual feedback after some trials and error, the virtual agents succeed to converge to a feasible behavior after several episodes even though it is a suboptimal path. Fig. 12c shows one episode with the two-step reward setting.

Another design choice could have been made to avoid diagonal movements, that is to model the environment so that it would not make the avatar move if the two virtual agents had chosen different actions and giving them a negative reward. Anyway, this choice would not have been consistent with the human experiment in which the sum of the two actions determines the avatar behavior. Nevertheless, the final results would not have been different from the one obtained with the two-step reward setting after 50000 episodes.

5. Conclusion

This work proposes a two-fold study investigating the cognitive collaboration process both among humans and virtual agents of a multi-agent reinforcement learning (MARL) system.

The first is a system offering an interactive virtual shared environment for experimenting with cognitive collaboration using multimodal channels (visual and haptics). The implemented experiment proposes a task in which the users should escape a maze, trying to collect the best score possible. The task has to be done in pair since the visual information provided to the participants were split: one of the participants could see the wall of the labyrinth and the goal position while the other could perceive the bombs and the bonuses. Two large workspace haptic interfaces have been used to move the camera in the scene. The final movement of the virtual player is determined by the relative motion of the two haptic devices. The partners perceive each own haptic feedback and not the force of the companion, can not see each other since they impersonate the same virtual agent (seeing the scene in a first view perspective) but can derive the intention of the other member observing how the virtual displacement differs from the generated forces. The experiments demonstrate that a multimodal channel consisting of vision and haptics allows establishing a common frame of reference (CRF) between the two participants for learning how to collaborate to accomplish a shared task.

The machine learning experiment proposes a new MARL model, which is tested in a simplified 2D grid maze with walls and bombs, and two virtual agents implemented as tabular Q-learning perceiving different and complementary cues of the environment necessary for an avatar to escape the maze. The combination of the two agents' actions determines the behavior of the avatar in the environment. According to the movement of the avatar, the two virtual agents receive two different rewards. Three different policies of action combination have been analyzed. The aim is to find out if both agents can learn the way to reach the goal unharmed, and therefore, to explore the possibility of a virtual agent to learn the intention of the other one. Only the last setting (two-step reward) shows satisfactory results demonstrating some form of learning among the agents with the avatar that performs feasible actions. Indeed, such a setting is the one that most resembles the human experiment. The second-step reward can be seen as the mental process of one participant to infer the intention of the other, seeing how much the movement of the avatar differs from the one that he/she controlled with the haptic device.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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